P. Praveen Kumar

(**Senior AI Engineer**)

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Professional Summary

- **Senior Data Scientist** and Software Developer with **8.4 years** of expertise in **Artificial Intelligence**.
- Highly skilled in **Generative AI**, **Computer Vision**, **Machine learning**, **Predictive Modelling**.
- Hands-on experience with **TensorFlow**, **Keras**, **PyTorch**, **PIL**, **Groq, RAGAS**, **LangSmith**.
- Specialized in **Agentic Workflows (LangGraph, CrewAI), Self-RAG ecosystem, State Management**.
- Experience with **TensorFlow Serving**, **Docker**, **AWS ECS/Fargate**, **GitHub Actions, MLflow**.
- Deep Knowledge of **Transformers**, **Hugging Face, BERT**, **Autoregressive models (GPT-4)**.
- Worked with **Microsoft Graph API**, **Heroku**, **MongoDB, Snowflake**, **FastAPI, GCP (Vertex AI)**.
- Proficient in **OpenCV, YOLO**, **GAN**, **VAE**, **Diffusion Models**, **LLaMA**, **state-of-the-art architectures**.
- Advanced **LLM fine-tuning** (**LORA/QLORA)**, **Prompt Engineering**, **Quantization (GGUF,AWQ)**.
- Experienced in **Core Java**, **J2EE**, **Spring**, **MuleSoft (API)**, **Feature Store (Feast)**.
- Proficient in **Ensemble Methods (Stacking, Boosting), PCA, Hypothesis Testing, ANOVA**.
- Skilled in **Deep learning**, **Natural Language Processing**, **Conversational AI, Graph DB (Neo4j)**.
- Competency in **Model Orchestration, Vector DBs, Microsoft Azure, AKS**.
- Specialized in **Text Analytics**, **Video Analytics**, **Object Detection**, **and Audio Analytics**.
- **Pro-degree in Data science**.

Work Experience

Synechron - Lead GEN AI Developer / Senior Data Scientist 10/2024 – 10/2025

Project: AI-Powered Outlook Report Generation System (Mashreq)

- Architected an autonomous multi-agent economic intelligence system that automates end-to-end generation of strategic reports (GCC Economic Insights & Global Monetary Outlook), ensuring analyst-grade accuracy.
- Engineered a hierarchical multi-agent workflow using LangGraph, featuring a Supervisor Agent that utilizes ReAct planning to orchestrate specialized sub-agents (Data Retrieval, Economic Insight, Monetary Policy) with shared state memory & conditional routing logic to ensure cross-sectional narrative generation.
- Developed a robust Data Retrieval Agent with custom tools to aggregate and normalize macroeconomic indicators from Bloomberg, IMF, and OPEC, integrating a Code Interpreter to perform real-time statistical analysis, dynamically generate visualizations (GDP heatmaps, inflation variance, trend decomposition).
- Implemented a router-based agentic RAG, dynamically selecting Llama 3 for low-latency reasoning and Claude 3.5 Sonnet via AWS Bedrock for high-precision final drafting, and E5-Mistral for semantic retrieval.
- Deployed a human-in-the-loop evaluation pipeline using an LLM-as-a-Judge (self-correction) to validate report accuracy against ground-truth financial data, ensuring zero-hallucination for critical monetary reports.

Project: Internal Policy Assistant

- Architected and deployed an enterprise-grade, security-first RAG ecosystem for internal compliance using Azure OpenAI (GPT-4o) via Private Endpoints, for instant Q&A over sensitive policy documents.
- Engineered a production-grade hybrid search pipeline on PostgreSQL (pgvector), combining dense retrieval (bge-large-en-v1.5), sparse keyword search (BM25) & cross-encoder re-ranking, achieving 90%+ retrieval precision on complex legal queries. Experimented with Llama3(Groq), Claude 3.5 Sonnet, Falcon 180B, Mixtral-8x22B & embedding models like text-embedding-3-large, E5-Mistral.
- Implemented a Self-RAG framework using a specialized Small Language Model (Google Gemma 2) as a validation judge to critique GPT-4o response fidelity, significantly reducing hallucinations and improving factual consistency, logging rejections to construct DPO datasets for model fine-tuning, while leveraging Azure AI Content Safety Guardrails for low-latency input/output filtering.
- Orchestrated containerized embedding services on Azure Kubernetes Service (AKS), delivering sub-200ms p95 vectorization latency with LangSmith-based prompt, pipeline, and version tracking.

Project: Smart Waste Management System - DU Client

- Designed and implemented an AI-powered Smart Waste Management solution for DU client to enhance waste segregation efficiency and sustainability awareness in airport environments. Leveraged a YOLOv8 (Pytorch) model integrated with OpenCV for real-time video analysis and TensorRT for optimized edge inference, accurately classifying biodegradable, plastic/non-biodegradable & metal/glass waste items.
- Developed an end-to-end edge deployment pipeline on Raspberry Pi with integrated camera. The model accurately detects the waste through video analysis. Applied bounding box-level augmentations (flip, hue, grayscale, Mixup) along with frame extraction, resizing, and brightness normalization using OpenCV. Leveraged transfer learning and non-maximum suppression for robust waste detection.
- Integrated Azure IoT Hub and deployed backend APIs on Azure, enabling IoT connectivity, telemetry monitoring. Selected Model based on Precision, Recall, IOU, box AP, mAP50, mAP50-95, TPR, FPR.
- Containerized model inference using Docker & orchestrated workloads with Kubernetes on Azure, managed model versioning with MLflow Model Registry, utilized Cosmos DB for telemetry & storage.

Project: Mashreq Virtual Banking Assistant

- Designed and implemented a GenAI-powered multilingual virtual assistant for Mashreq Bank to assist customers in Arabic and English, providing real-time guidance on loans, mutual funds & personalized financial recommendations. Built Agentic RAG pipeline with document ingestion, preprocessing, chunking, and indexing using Azure Cognitive Search for context-aware and compliant responses.
- Integrated Whisper Large-v3 for multilingual Speech-to-Text conversion with real-time streaming transcription. Deployed as a containerized microservice within the Azure environment, enabling seamless voice-based interaction and conversation persistence across sessions.
- Deployed Llama 3 on Azure GPU infrastructure using Kubernetes for orchestration and MLflow Model Registry for versioning. Served through REST/gRPC inference APIs with autoscaling & monitoring.
- Utilized privacy-safe embedding models (bge-large-en-v1.5, E5-Mistral) hosted in a private cloud for multilingual vector generation. Stored embeddings in a FAISS/Azure Cognitive Search vector index.
- Integrated Guardrails AI for content safety, data masking, and regulatory compliance enforcement. Implemented a policy-aware response layer to detect & block disallowed & sensitive contents.
- Delivered an end-to-end GenAI microservice architecture, fully containerized with Docker, deployed on Azure Kubernetes Service (AKS) with Azure API Management Gateway for secure routing, Application Insights & Prometheus for observability, and Azure AD authentication.

InfoVision –Senior Data Scientist (Technical Specialist) **03/2023 – 08/2024**

Project: Titan 3/Verizon 5G Home Internet

- Designed and implemented a camera-based detection system within a mobile application to capture and analyze the LED indicators' statuses in real-time. This system empowers users in the autonomous self-installation of 5G routers, eliminating the need for customer support. Techniques included frame extraction, letterbox resizing, and contrast enhancement using OpenCV for video processing.
- Developed a robust YOLOv8 model by applying transfer learning to analyze real-time camera feed data, utilizing non-maximum suppression to filter overlapping bounding boxes and rescaling predictions. The model accurately detects the router LED light statuses, such as signal strength and Wi-Fi indicators, through video analysis, delivering actionable insights to users.
- Integrated motion detection algorithms like contour-based methods to detect significant motion changes in LED signal indicators, comparing consecutive video frames. Managed model versioning and deployment using MLflow Model Registry, ensuring efficient model tracking and organization.
- Designed scalable backend APIs for seamless data transmission and model inference, leveraging AWS GPU machines for efficient model training and deployment.

Project: Automotive Surface Anomaly Detection System

- Developed a mobile application facilitating real-time detection of car dents and scratches, allowing users to authenticate, upload car images/videos, and obtain severity analysis results from backend model. Implemented mongo database for efficient user history tracking and record management.
- Worked with bounding box-level augmentations (flip, hue, saturation, grayscale, MixUp,and CutMix), image-level augmentations (90° rotation, horizontal flip, and Mosaic).
- Evaluated Detectron2, YOLOv8, and Mask R-CNN for dent detection, selecting the most suitable model based on accuracy metrics (Precision, Recall, F1-score, IOU), detection performance (box AP, mAP50, mAP50-95, TPR, FPR), and model efficiency (inference time, model size).
- Deployed the optimized model & backend inference service on AWS EC2 with GPU acceleration, ensuring low-latency prediction and high availability, & integrated S3 for media storage & retrieval

Project: HealthCareQAassist – ORIX India

- Leveraged advanced natural language processing to interact with user personas, including patients and hospital staff, providing tailored responses based on their specific profiles. The system efficiently retrieves and processes relevant medical information.
- Integrated Faiss for efficient vector storage and retrieval within a Retrieval-Augmented Generation (RAG) technique, utilizing Azure OpenAI's GPT-4 to generate context-aware responses from the retrieved data. LangChain manages multi-step reasoning via chains, while agentic systems handle dynamic queries, ensuring seamless information retrieval, processing, and real-time response generation.
- The model effectively addresses patient queries about treatment options or appointment scheduling, providing context-specific information based on medical history or preferred healthcare providers. Langsmith was employed to track different versions of prompts, chains, and agents.
- Worked with HuggingFace models like Mistral-8x7B, Zephyr 7b, Meta's Llama2 70b, Llama 3, along with Qwen2 and evaluated them using Ragas, perplexity, BERTScore, HumanEval.
- Experimenting with latest LLMs like GPT-4o, PALM2, Gemini Pro, Stable Zephr as well as embedding models like text-embedding-ada-2, sentence-transformers/all-MiniLM-L6-v2 and NLP topics such as tree parsing, semantic analysis, voice assistants, transcription services, deeper linguistic understanding (CFG/CNF).

POC: Henry Schein Propensity Modeling

- Custom propensity model that predicts the likelihood of a customer placing an order for a particular product in the following week.
- RFM (Recency, Frequency and Monetary) analysis and customer lifetime value functions are further used to identify the frequent customers to obtain a total of 300K item-customer combinations.

Hexaware - Senior Data Scientist 11/2021 – 01/2023

Project: Thriller/JOSH

- Built a Customized Human Action Recognition Video Classification Model using Alexnet (State of Art) with Transfer Learning techniques.
- CNN Model that detects NSFW (not safe for work) activities in the video and classifies the video depending on the content as safe or unsafe.
- Robust Video Classification method, such as CNN with LSTM is applied, which makes use of both spatial and temporal information in a video. Both are used for learning features extraction and predicting the associated movements. Moving/Rolling averaging technique is used to avoid flickering effectively.
- Transfer Learning technique is used with customized 5(NSFW) Categories.
- AssemblyAI API is used to identify any profanity words in the videos. FFMPEG is used for extracting the audio from the video, extracted audio is given to AssemblyAI for Analysis. AssemblyAI transcribes the audio into text format and helps to find any profanity is used in the whole audio.

Project: Claim Processing/HDFC

- Claim Processing helps to classify the Assurant email based on the categories the particular mail belongs Tesseract, OCR and BeautifulSoup are used for extracting the texts from body, subject, attachment and image files received in Assurant mail and extracted texts are classified based on the categories.
- Models used are Extra Tree Classifier, Gradient Boosting Classifier, Voting Classifier.
- TF-IDF used for Vectorizing the input texts and ML model predicts the probability scores for all the categories and threshold value of 0.8 is set for classifying the category.
- Watch script which monitors the emails 24/7 for every 10 seconds.
- Flask API which uses the trained pickle model for responding to the prediction after Word Embedding.
- Actively engaged in quantitative analysis of sophisticated modelling to address business needs.

Project: Proctoring

- An AI based Proctoring Employee Monitoring tool, that can improve workforce management, flag unwanted events and increase transparency within an organization leveraging unbiased measurement system with an ability to customize rules/triggers for flagging. Keeps track of employee's productivity, compliance and security. Permits remote work distribution and allocation with efficiency.
- YoloV3 is used for object detection in monitoring screens.
- Flagging unwanted events such as an unidentified person trying to access user machine, using mobile phones during production hours and taking screenshots of company data on the screen, etc.
- Two-factor authentication with user ID, Password and facial data to avoid fake/unidentified logins.
- View the violations and escalate them based on pre-defined categories. Employee's facial data are stored in MongoDB. For Face recognition, Dlib library is used.

Project: Contact Centre

- Converting speech of the agent and customer to text using google speech to text Speech Recognition API service. Based on the conversation, model predicts the category which is needed for the customer.
- Prebuild and set of rules for the bot to respond. Eg: bank account details.

TCS – Data Scientist (Payroll – Crown Solutions) **09/2020 – 09/2021**

Project: Hartford Insurance

- Worked in creating a customized python package (automata) which helps to extract, transform and validate company's internal data and has features like storing the model metadata into oracle and snowflake tables using PySpark.
- Worked with data filtering, EDA process and CI/CD pipeline.
- Visualizing the insights of the customer's policy/insurance data for every quarter.
- CI Engine which is used to analyses and compare the quotes data of the companies to give better policy rate for the customers.
- Automated the scripts using AutoSys which collects the latest data from hive and generates reports.
- Actively engaged in quantitative analysis of sophisticated modelling to address business needs.
- Worked on Virtual Environment for creating packages and customized kernels.
- Selecting the better model using MLFlow and Model Registry

TCS – Data Scientist (Payroll – Crown Solutions) **07/2018 – 07/2019**

Project: SPRINT DIGITAL TELECOM (SERVICE PLATFORM)

- Analyzing the Storage space usage for snowflake services for every quarter.
- Approached **Time Series analysis** which helps to identify the trends in our compute habits.
- Trend Analysis for days/times when resources are used infrequently and limit the compute nodes during slow times. Drill down into databases, schemas and query types to track adoption across the organization with user adoption.
- Analyzing the monthly average for each individual project and usage over time which lies under cost per limitation of business.
- Tuning the predictive churn user activity from past and detecting which customers are likely to cancel the subscription to a service based on their usage. **Two class Logistic** and **Boosted Decision Tree**.

Geetham Software Pvt Ltd – Junior Data Scientist **07/2017 – 06/2018**
Project: TIME Dataset / Aadhaar Dataset

- Collected, studied and interpreted large datasets from different sources for the company and cleansed it for further analysis.
- Analyzed the sales trends of the company and helped to predict their requirements for the upcoming sales. Reported healthy and unhealthy months of the company in order to improve their revenue.
- Suggested to determine what actions to take for targeting the right area of customers in Aadhaar.
- Packages used are **Numpy, pandas, Matplotlib, sklearn, Keras.**
- Developed and implemented new forecasting models which increased company productivity and efficiency.
- Implemented Sentimental Analysis for Aadhaar using NLP (NLTK).
- Visualized the data using Tableau. Also got customer appreciations for the reports.

Artificial Neural Networks
Developed a POC model in **ANN.**
Based on the track of six months of the randomly picked customers, have collected those customers who left and stayed in the bank.
Predictive model which will say which of their customers are at highest risk of leaving the bank.

Geetham Software Pvt Ltd – MuleSoft Developer **01/2016 – 06/2017**
Project: Transport Administration (TIME-Transport Income Made Easy)

- Worked on development of RAML 0.8 using Anypoint exchange.
- Integrating the layer for the Module application using Mule Soft.
- Developed flow for a set of messages to be converted into the canonical format.
- MULE EXPRESSION LANGUAGE (MEL) is used for development.
- Continuous integration and continuous development using Jenkins.
- Implemented development of Error framework for the service layer.
- Worked in various connectors, transformers and Listeners.
- Bugging out the developed modules using Munit test cases for the integrations.
- Profoundly worked on DATAWEAVE EXPRESSION Language.

Skills

Programming Language	Python, Java, Spring, MuleSoft
Artificial Intelligence/DL	Computer Vision, Yolo, Darknet, Transformers, Gen AI, Llama3, LLMs, RAG, GANs, VAEs, GPT3/4, LORA, BERT, HuggingFace models, Deep Learning, NLP, Machine Learning, OpenAI, AI Agents
Data Analytics	R, PySpark, Plotly, Bokeh, Altair, sanky, sunburst
Databases	MySQL, Oracle, SnowFlake, Hive, MongoDB, Knowledge Graph
IDE	Eclipse, Anypoint Studio, Jupyter Notebook, Spyder, Colab, VS code
Frameworks/Packages	Beautifulsoup, Pytesseract, OpenCV, Pytorch, NLTK, Sklearn, LangGraph, Llamaindex, langsmith
Repository	Git, Bitbucket, MLFlow, Github actions
Operating Systems	Windows, Unix/Linux
Tools	Tableau, Excel, WinSCP, Putty, Docker, Dlib, Streamlit
Deployment	AWS EC2/ECS, Heroku, Microsoft Graph API, Feature store(Feast), AWS Bedrock, Vertex AI
Project Tracking Tools	JIRA, Rally
Web Services	RESTful, Postman, Flask, FastAPI, Microsoft Azure, GCP

Education

St. Joseph's College of Engineering Chennai	**Electronics and Communication Engineering** Graduated with Aggregate of 56%	**09/2011 - 06/2015**
Vidhya Vikas Higher Sec School	**12th – State Board** Graduated with Aggregate of 70%	**09/2009 - 06/2011**
Petit Seminaire Pondicherry	**10th – State Board** Graduated with Aggregate of 77%	**09/1997 - 06/2009**

Profile

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Date: P. Praveen Kumar